



OFFICIAL OLYMPIAD SYLLABUS

Classes 5 to 12

Mathematics • Science • Physics • Chemistry • Biology

PURPOSE

A school-aligned, concept-oriented syllabus designed to define what students may be assessed on in the National Genius Olympiad.

APPROACH

School concepts first. Olympiad difficulty comes primarily from reasoning, application, interpretation and problem solving—not from university-level content.

CURRICULUM BASIS

The framework is normalized from common Indian school curriculum patterns, using the supplied CBSE/ICSE comparison material as the primary working source.

1. About the National Genius Olympiad

The National Genius Olympiad syllabus is designed to provide a clear and progressive academic framework for students from Classes 5 to 12. It identifies the major concepts normally encountered in Indian schools and organizes them into a consistent Olympiad structure.

The syllabus is intentionally different from a textbook contents page. Chapters are organized around concepts so that the same idea can be introduced at an appropriate age, strengthened in later classes and eventually used in higher-order applications.

Core principle: Students should be challenged by how they use a concept, not by being required to study an unrelated advanced topic.

M	S	P	C	B
Mathematics	Science	Physics	Chemistry	Biology

2. Syllabus Policy

- The syllabus is primarily aligned to concepts commonly taught in Indian schools, with common CBSE/NCERT and other major-board patterns used to determine progression.
- Board sequencing is not identical. Where a topic varies substantially by board or school, it may be treated as Board-dependent / Optional.
- The syllabus defines the assessable concept area. Questions may use unfamiliar contexts, diagrams, graphs, tables, experiments or multi-step situations.
- For Classes 5–10, the official Science subject remains Science even though concepts naturally draw from physics, chemistry, biology and environmental science.
- For Classes 11–12, Mathematics, Physics, Chemistry and Biology are separated into distinct senior-secondary syllabi.
- The document is an Olympiad syllabus framework and is not an official publication of NCERT, CBSE, CISCE or any other education board.

3. How to Read Each Chapter

Topics define the knowledge and concepts that may be assessed. **Olympiad application** may require students to reason with those concepts through comparisons, calculations, patterns, graphs, diagrams, real-world situations and multi-step problems.

4. Master Syllabus Overview

The overview below lists every class, subject, number of chapters and the page where that detailed syllabus section begins.

Class	Subject	Chapters	Page
Class 5	Mathematics	12	4
Class 5	Science	10	8
Class 6	Mathematics	14	12
Class 6	Science	12	16
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Class 11	Mathematics	13	60
Class 11	Physics	10	64
Class 11	Chemistry	9	67
Class 11	Biology	10	70
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Class 12	Chemistry	10	82
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CLASS 5

School-level foundation with progressive Olympiad application.

Mathematics

Focus: computational accuracy, patterns, algebraic thinking, geometry, data interpretation and logical problem solving.

Chapter 1 — Number Sense and Large Numbers

Topics

- Place value and face value
- Indian and International number systems
- Reading and writing large numbers
- Comparing and ordering numbers
- Expanded and standard forms
- Rounding and estimation
- Number lines
- Even and odd numbers
- Number patterns
- Real-life numerical information

Important concepts: Place value and face value, Indian and International number systems, Reading and writing large numbers, Comparing and ordering numbers.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Four Fundamental Operations

Topics

- Addition
- Subtraction
- Multiplication
- Division
- Order of operations
- Mental calculation
- Estimation
- Unit-based calculations
- Word problems
- Multi-step problems

Important concepts: Addition, Subtraction, Multiplication, Division.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Factors, Multiples and Number Patterns

Topics

- Factors
- Multiples
- Prime and composite numbers
- Divisibility
- Common factors
- Common multiples
- HCF
- LCM
- Prime factorisation
- Number puzzles

Important concepts: Factors, Multiples, Prime and composite numbers, Divisibility.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Fractions

Topics

- Proper, improper and mixed fractions
- Equivalent fractions
- Comparing fractions
- Fractions on the number line
- Addition and subtraction
- Multiplication and division
- Fraction as quantity
- Fraction as division
- Applications

Important concepts: Proper, improper and mixed fractions, Equivalent fractions, Comparing fractions, Fractions on the number line.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — Decimals

Topics

- Decimal place value
- Reading and comparing decimals
- Ordering decimals
- Decimal-fraction relationship
- Addition and subtraction
- Multiplication and division
- Money and measurement applications

Important concepts: Decimal place value, Reading and comparing decimals, Ordering decimals, Decimal-fraction relationship.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Ratio, Proportion and Unitary Thinking

Topics

- Meaning of ratio
- Writing and comparing ratios
- Equivalent ratios
- Simple proportion
- Unitary method
- Scaling quantities
- Real-life applications

Important concepts: Meaning of ratio, Writing and comparing ratios, Equivalent ratios, Simple proportion.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Percentage and Everyday Mathematics

Topics

- Meaning of percentage
- Percentage as fraction
- Fraction-to-percentage conversion
- Percentage-to-decimal conversion
- Simple percentage problems
- Comparing quantities
- Money applications
- Simple discounts

Important concepts: Meaning of percentage, Percentage as fraction, Fraction-to-percentage conversion, Percentage-to-decimal conversion.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Basic Algebra and Patterns

Topics

- Number and shape patterns
- Finding a pattern rule
- Unknown quantities
- Simple expressions
- Variables as unknowns
- Simple equations
- Substitution

Important concepts: Number and shape patterns, Finding a pattern rule, Unknown quantities, Simple expressions.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Geometry: Lines, Angles and Shapes

Topics

- Points, lines, segments and rays
- Parallel and intersecting lines
- Acute, right, obtuse and straight angles
- Triangles
- Quadrilaterals
- Polygons
- Basic geometric reasoning

Important concepts: Points, lines, segments and rays, Parallel and intersecting lines, Acute, right, obtuse and straight angles, Triangles.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 10 — Symmetry, Patterns and Spatial Reasoning

Topics

- Line symmetry
- Reflection
- Symmetric figures
- Geometric patterns
- Tessellations
- Shape arrangements
- Visualisation
- Spatial reasoning

Important concepts: Line symmetry, Reflection, Symmetric figures, Geometric patterns.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 11 — Measurement and Mensuration

Topics

- Length
- Mass
- Capacity
- Time
- Unit conversion
- Perimeter
- Area
- Volume/capacity
- Estimation
- Practical measurement

Important concepts: Length, Mass, Capacity, Time.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 12 — Data Handling and Logical Mathematics

Topics

- Tables
- Pictographs
- Bar graphs
- Simple line representations
- Comparing data
- Patterns
- Number puzzles
- Logical reasoning
- Problem solving

Important concepts: Tables, Pictographs, Bar graphs, Simple line representations.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Science

Focus: conceptual science, observation, cause-and-effect reasoning, diagrams, experiments, applications and interpretation.

Chapter 1 — Scientific Thinking and Investigation

Topics

- Observation
- Asking questions
- Prediction
- Investigation
- Evidence
- Drawing conclusions
- Measurement
- Scientific safety

Important concepts: Observation, Asking questions, Prediction, Investigation.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Water and Its Importance

Topics

- Sources of water
- Uses of water
- Water and living organisms
- Water requirements
- Water conservation
- Water pollution
- Responsible use

Important concepts: Sources of water, Uses of water, Water and living organisms, Water requirements.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Food, Nutrition and Health

Topics

- Sources of food
- Plant and animal foods
- Nutrients
- Carbohydrates
- Proteins
- Fats
- Vitamins
- Minerals
- Balanced diet
- Hygiene

Important concepts: Sources of food, Plant and animal foods, Nutrients, Carbohydrates.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Living Organisms

Topics

- Living and non-living things
- Characteristics of life
- Plants
- Animals
- Habitats
- Adaptation
- Biodiversity
- Interdependence

Important concepts: Living and non-living things, Characteristics of life, Plants, Animals.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — Matter and Materials

Topics

- Materials around us
- Properties of materials
- Solids, liquids and gases
- Solubility
- Transparent, translucent and opaque materials
- Hard and soft materials
- Useful and harmful materials

Important concepts: Materials around us, Properties of materials, Solids, liquids and gases, Solubility.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Separation of Materials

Topics

- Mixtures
- Handpicking
- Sorting
- Sieving
- Winnowing
- Sedimentation
- Decantation
- Filtration
- Evaporation
- Choosing a separation method

Important concepts: Mixtures, Handpicking, Sorting, Sieving.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Changes Around Us

Topics

- Physical changes
- Reversible and irreversible changes
- Heating and cooling
- Melting
- Freezing
- Evaporation
- Condensation
- Useful and harmful changes

Important concepts: Physical changes, Reversible and irreversible changes, Heating and cooling, Melting.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Motion and Measurement

Topics

- Distance
- Measurement
- Standard units
- Simple motion
- Rest and movement
- Introductory speed
- Measuring instruments

Important concepts: Distance, Measurement, Standard units, Simple motion.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Light, Sound and Shadows

Topics

- Sources of light
- Transparent materials
- Shadows
- Formation and size of shadows
- Reflection
- Sound production
- Vibration
- Loud and soft sounds
- Musical instruments

Important concepts: Sources of light, Transparent materials, Shadows, Formation and size of shadows.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 10 — Earth and Environment

Topics

- Earth as a home
- Land
- Water
- Air
- Natural resources
- Plants and animals
- Waste
- Pollution
- Conservation
- Human impact

Important concepts: Earth as a home, Land, Water, Air.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

CLASS 6

School-level foundation with progressive Olympiad application.

Mathematics

Focus: computational accuracy, patterns, algebraic thinking, geometry, data interpretation and logical problem solving.

Chapter 1 — Number Systems

Topics

- Natural and whole numbers
- Number line
- Successor and predecessor
- Properties of operations
- Large numbers
- Indian and International systems
- Estimation

Important concepts: Natural and whole numbers, Number line, Successor and predecessor, Properties of operations.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Integers

Topics

- Positive and negative integers
- Zero
- Number line
- Comparison
- Addition
- Subtraction
- Multiplication
- Division
- Integer applications

Important concepts: Positive and negative integers, Zero, Number line, Comparison.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Fractions and Decimals

Topics

- Equivalent fractions
- Comparing fractions
- Fraction operations
- Fraction as operator
- Decimal representation
- Decimal operations
- Fraction-decimal conversion
- Word problems

Important concepts: Equivalent fractions, Comparing fractions, Fraction operations, Fraction as operator.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Factors and Multiples

Topics

- Prime numbers
- Composite numbers
- Prime factorisation
- HCF
- LCM
- Divisibility rules
- Number relationships

Important concepts: Prime numbers, Composite numbers, Prime factorisation, HCF.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — Ratio and Proportion

Topics

- Ratio
- Equivalent ratios
- Comparison
- Proportion
- Unitary method
- Direct proportion
- Applications

Important concepts: Ratio, Equivalent ratios, Comparison, Proportion.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Algebraic Expressions

Topics

- Variables
- Constants
- Terms
- Coefficients
- Like and unlike terms
- Expressions
- Simplification
- Substitution

Important concepts: Variables, Constants, Terms, Coefficients.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Simple Equations

Topics

- Unknowns
- Equality
- One-variable equations
- Solving equations
- Word problems
- Checking solutions

Important concepts: Unknowns, Equality, One-variable equations, Solving equations.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Lines and Angles

Topics

- Point
- Line
- Segment
- Ray
- Parallel lines
- Intersecting lines
- Angles
- Angle relationships

Important concepts: Point, Line, Segment, Ray.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Triangles and Quadrilaterals

Topics

- Types of triangles
- Sides and angles
- Triangle properties
- Quadrilaterals
- Parallelogram
- Rectangle
- Square
- Rhombus
- Trapezium

Important concepts: Types of triangles, Sides and angles, Triangle properties, Quadrilaterals.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 10 — Symmetry and Constructions

Topics

- Line symmetry
- Reflection
- Symmetric figures
- Geometric constructions
- Compass and ruler
- Patterns and tessellations

Important concepts: Line symmetry, Reflection, Symmetric figures, Geometric constructions.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 11 — Perimeter and Area

Topics

- Perimeter
- Area
- Rectangle
- Square
- Triangle
- Composite figures
- Unit conversion
- Practical problems

Important concepts: Perimeter, Area, Rectangle, Square.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 12 — Mensuration and Volume

Topics

- Cube
- Cuboid
- Volume
- Capacity
- Surface-based reasoning
- Unit conversion

Important concepts: Cube, Cuboid, Volume, Capacity.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 13 — Data Handling

Topics

- Data collection
- Tables
- Pictographs
- Bar graphs
- Mean
- Median
- Mode
- Interpretation

Important concepts: Data collection, Tables, Pictographs, Bar graphs.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 14 — Patterns and Logical Reasoning

Topics

- Number patterns
- Sequences
- Odd/even patterns
- Number puzzles
- Cryptarithms
- Logical relationships
- Mathematical reasoning

Important concepts: Number patterns, Sequences, Odd/even patterns, Number puzzles.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Science

Focus: conceptual science, observation, cause-and-effect reasoning, diagrams, experiments, applications and interpretation.

Chapter 1 — Nature of Science and Measurement

Topics

- Scientific observation
- Questions and hypotheses
- Evidence
- Experiments
- Measurement
- Units
- Scientific safety

Important concepts: Scientific observation, Questions and hypotheses, Evidence, Experiments.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Matter and Its Properties

Topics

- Matter
- Particles
- Solids
- Liquids
- Gases
- Particle arrangement
- Inter-particle space
- Change of state

Important concepts: Matter, Particles, Solids, Liquids.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Separation of Substances

Topics

- Mixtures
- Solubility
- Filtration
- Sedimentation
- Decantation
- Evaporation
- Separation based on properties

Important concepts: Mixtures, Solubility, Filtration, Sedimentation.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Living World and Classification

Topics

- Living organisms
- Characteristics of life
- Classification
- Plants
- Animals
- Habitats
- Adaptation
- Biodiversity

Important concepts: Living organisms, Characteristics of life, Classification, Plants.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — Food and Nutrition

Topics

- Food sources
- Nutrients
- Balanced diet
- Deficiency
- Digestion
- Food safety

Important concepts: Food sources, Nutrients, Balanced diet, Deficiency.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Body Movement

Topics

- Human skeleton
- Bones
- Joints
- Muscles
- Movement
- Animal movement

Important concepts: Human skeleton, Bones, Joints, Muscles.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Measurement and Motion

Topics

- Length
- Mass
- Time
- Distance
- Motion
- Speed
- Simple motion graphs

Important concepts: Length, Mass, Time, Distance.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Light and Shadows

Topics

- Sources of light
- Transparent materials
- Shadows
- Reflection
- Plane mirrors
- Image formation

Important concepts: Sources of light, Transparent materials, Shadows, Reflection.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Electricity and Circuits

Topics

- Electric current
- Cells
- Batteries
- Circuit
- Conductors
- Insulators
- Switches
- Bulbs
- Series connections

Important concepts: Electric current, Cells, Batteries, Circuit.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 10 — Magnets

Topics

- Magnetic materials
- Magnetic poles
- Attraction
- Repulsion
- Compass
- Introductory magnetic field

Important concepts: Magnetic materials, Magnetic poles, Attraction, Repulsion.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 11 — Heat and Temperature

Topics

- Hot and cold
- Temperature
- Thermometer
- Celsius scale
- Heating
- Cooling
- Basic heat transfer

Important concepts: Hot and cold, Temperature, Thermometer, Celsius scale.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 12 — Water, Air and Environment

Topics

- Water cycle
- Air
- Atmosphere
- Natural resources
- Pollution
- Conservation
- Waste management

Important concepts: Water cycle, Air, Atmosphere, Natural resources.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

CLASS 7

School-level foundation with progressive Olympiad application.

Mathematics

Focus: computational accuracy, patterns, algebraic thinking, geometry, data interpretation and logical problem solving.

Chapter 1 — Integers and Rational Numbers

Topics

- Integer operations
- Properties
- Rational numbers
- Number line
- Comparison
- Rational-number operations
- Word problems

Important concepts: Integer operations, Properties, Rational numbers, Number line.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Fractions and Decimals

Topics

- Operations
- Fraction as operator
- Decimal multiplication
- Decimal division
- Applications

Important concepts: Operations, Fraction as operator, Decimal multiplication, Decimal division.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Powers and Exponents

Topics

- Powers
- Base and exponent
- Laws of exponents
- Repeated multiplication
- Standard form
- Large-number representation

Important concepts: Powers, Base and exponent, Laws of exponents, Repeated multiplication.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Algebraic Expressions

Topics

- Variables
- Constants
- Terms
- Coefficients
- Like terms
- Simplification
- Addition and subtraction of expressions

Important concepts: Variables, Constants, Terms, Coefficients.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — Linear Equations

Topics

- One-variable equations
- Transposition
- Equation solving
- Word problems
- Verification

Important concepts: One-variable equations, Transposition, Equation solving, Word problems.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Ratio, Proportion and Unitary Method

Topics

- Ratios
- Proportions
- Direct proportion
- Inverse proportion
- Unitary method
- Applications

Important concepts: Ratios, Proportions, Direct proportion, Inverse proportion.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Percentage and Comparing Quantities

Topics

- Percentage
- Fraction/decimal/percentage conversion
- Profit and loss
- Discount
- Simple interest
- Comparison of quantities

Important concepts: Percentage, Fraction/decimal/percentage conversion, Profit and loss, Discount.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Lines and Angles

Topics

- Parallel lines
- Transversal
- Corresponding angles
- Alternate angles
- Interior angles
- Angle relationships

Important concepts: Parallel lines, Transversal, Corresponding angles, Alternate angles.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Triangles

Topics

- Classification
- Angle properties
- Side-angle relationships
- Congruence concepts
- Construction-based reasoning

Important concepts: Classification, Angle properties, Side-angle relationships, Congruence concepts.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 10 — Congruence and Symmetry

Topics

- Congruent figures
- SSS
- SAS
- ASA/AAS
- RHS
- Symmetry
- Reflection

Important concepts: Congruent figures, SSS, SAS, ASA/AAS.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 11 — Perimeter and Area

Topics

- Triangle
- Quadrilateral
- Circle
- Composite shapes
- Practical applications

Important concepts: Triangle, Quadrilateral, Circle, Composite shapes.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 12 — Data Handling

Topics

- Data collection
- Mean
- Median
- Mode
- Bar graphs
- Double bar graphs
- Introductory probability

Important concepts: Data collection, Mean, Median, Mode.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 13 — Visualising Solid Shapes

Topics

- 2-D and 3-D shapes
- Faces
- Edges
- Vertices
- Nets
- Views
- Spatial reasoning

Important concepts: 2-D and 3-D shapes, Faces, Edges, Vertices.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 14 — Patterns and Logical Problem Solving

Topics

- Number patterns
- Algebraic patterns
- Geometric patterns
- Logical puzzles
- Mathematical reasoning

Important concepts: Number patterns, Algebraic patterns, Geometric patterns, Logical puzzles.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Science

Focus: conceptual science, observation, cause-and-effect reasoning, diagrams, experiments, applications and interpretation.

Chapter 1 — Nutrition in Plants

Topics

- Photosynthesis
- Chlorophyll
- Sunlight
- Carbon dioxide
- Water
- Starch
- Plant nutrition

Important concepts: Photosynthesis, Chlorophyll, Sunlight, Carbon dioxide.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Nutrition in Animals

Topics

- Digestive system
- Digestion
- Absorption
- Assimilation
- Human nutrition

Important concepts: Digestive system, Digestion, Absorption, Assimilation.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Fibre and Materials

Topics

- Natural fibres
- Synthetic fibres
- Properties
- Uses
- Environmental considerations

Important concepts: Natural fibres, Synthetic fibres, Properties, Uses.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Heat

Topics

- Temperature
- Heat
- Thermometers
- Conduction
- Convection
- Radiation
- Everyday applications

Important concepts: Temperature, Heat, Thermometers, Conduction.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — Acids, Bases and Salts

Topics

- Acids
- Bases
- Indicators
- Natural indicators
- Neutral substances
- Neutralisation
- Everyday applications

Important concepts: Acids, Bases, Indicators, Natural indicators.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Physical and Chemical Changes

Topics

- Physical changes
- Chemical changes
- Reversible/irreversible changes
- Rusting
- Combustion
- Evidence of chemical change

Important concepts: Physical changes, Chemical changes, Reversible/irreversible changes, Rusting.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Motion and Time

Topics

- Distance
- Time
- Speed
- Uniform motion
- Non-uniform motion
- Simple graphs

Important concepts: Distance, Time, Speed, Uniform motion.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Electric Current

Topics

- Circuits
- Current
- Cells
- Batteries
- Conductors
- Insulators
- Electrical safety

Important concepts: Circuits, Current, Cells, Batteries.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Light

Topics

- Reflection
- Plane mirrors
- Image characteristics
- Multiple reflections
- Shadows

Important concepts: Reflection, Plane mirrors, Image characteristics, Multiple reflections.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 10 — Winds, Weather and Pressure

Topics

- Air pressure
- Winds
- Heating of air
- Cyclones
- Weather patterns

Important concepts: Air pressure, Winds, Heating of air, Cyclones.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 11 — Living Organisms and Reproduction

Topics

- Plant reproduction
- Pollination
- Fertilisation
- Seed formation
- Animal reproduction
- Life cycles

Important concepts: Plant reproduction, Pollination, Fertilisation, Seed formation.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 12 — Environment and Natural Resources

Topics

- Ecosystems
- Food chains
- Soil
- Water
- Forests
- Conservation
- Human impact

Important concepts: Ecosystems, Food chains, Soil, Water.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

CLASS 8

School-level foundation with progressive Olympiad application.

Mathematics

Focus: computational accuracy, patterns, algebraic thinking, geometry, data interpretation and logical problem solving.

Chapter 1 — Rational Numbers

Topics

- Rational numbers
- Number line
- Equivalent forms
- Operations
- Properties
- Applications

Important concepts: Rational numbers, Number line, Equivalent forms, Operations.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Exponents and Powers

Topics

- Integral powers
- Laws of exponents
- Standard form
- Scientific notation
- Large and small quantities

Important concepts: Integral powers, Laws of exponents, Standard form, Scientific notation.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Algebraic Expressions and Identities

Topics

- Terms
- Coefficients
- Like terms
- Addition/subtraction
- Multiplication
- Algebraic identities
- Factorisation

Important concepts: Terms, Coefficients, Like terms, Addition/subtraction.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Linear Equations

Topics

- Equations in one variable
- Simplification
- Transposition
- Word problems
- Mathematical modelling

Important concepts: Equations in one variable, Simplification, Transposition, Word problems.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — Linear Equations in Two Variables

Topics

- Ordered pairs
- Coordinate plane
- Linear relationships
- Tables
- Graphs
- Interpretation

Important concepts: Ordered pairs, Coordinate plane, Linear relationships, Tables.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Direct and Inverse Proportion

Topics

- Direct variation
- Inverse variation
- Proportional relationships
- Graphical interpretation
- Applications

Important concepts: Direct variation, Inverse variation, Proportional relationships, Graphical interpretation.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Comparing Quantities

Topics

- Percentage
- Profit and loss
- Discount
- Simple interest
- Compound interest
- Growth and depreciation

Important concepts: Percentage, Profit and loss, Discount, Simple interest.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Squares, Cubes and Roots

Topics

- Square numbers
- Cube numbers
- Square roots
- Cube roots
- Perfect squares/cubes
- Factor methods

Important concepts: Square numbers, Cube numbers, Square roots, Cube roots.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Geometry and Quadrilaterals

Topics

- Polygons
- Quadrilaterals
- Parallelogram
- Rectangle
- Rhombus
- Square
- Trapezium
- Angle relationships

Important concepts: Polygons, Quadrilaterals, Parallelogram, Rectangle.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 10 — Triangles and Pythagorean Reasoning

Topics

- Triangle properties
- Pythagorean theorem
- Converse
- Right triangles
- Applications

Important concepts: Triangle properties, Pythagorean theorem, Converse, Right triangles.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 11 — Circles

Topics

- Circle terminology
- Radius
- Diameter
- Chord
- Arc
- Circumference
- Basic circle properties

Important concepts: Circle terminology, Radius, Diameter, Chord.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 12 — Mensuration

Topics

- Area
- Surface area
- Volume
- Cuboid
- Cube
- Cylinder
- Composite solids

Important concepts: Area, Surface area, Volume, Cuboid.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 13 — Coordinate Geometry

Topics

- Coordinate plane
- Coordinates
- Plotting
- Distance concept
- Midpoint concept
- Geometric interpretation

Important concepts: Coordinate plane, Coordinates, Plotting, Distance concept.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 14 — Data Handling and Probability

Topics

- Data collection
- Frequency
- Tables
- Graphs
- Mean
- Median
- Mode
- Basic probability

Important concepts: Data collection, Frequency, Tables, Graphs.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 15 — Patterns and Mathematical Reasoning

Topics

- Sequences
- Number patterns
- Algebraic patterns
- Logical puzzles
- Visual reasoning
- Problem solving

Important concepts: Sequences, Number patterns, Algebraic patterns, Logical puzzles.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Science

Focus: conceptual science, observation, cause-and-effect reasoning, diagrams, experiments, applications and interpretation.

Chapter 1 — Matter and Its Structure

Topics

- Particle nature of matter
- Atoms and molecules
- States of matter
- Inter-particle space
- Kinetic ideas
- Changes of state

Important concepts: Particle nature of matter, Atoms and molecules, States of matter, Inter-particle space.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Physical and Chemical Changes

Topics

- Physical changes
- Chemical changes
- Reactions
- Rusting
- Combustion
- Energy changes

Important concepts: Physical changes, Chemical changes, Reactions, Rusting.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Materials and Metals

Topics

- Metals
- Non-metals
- Physical properties
- Chemical properties
- Uses
- Corrosion
- Prevention

Important concepts: Metals, Non-metals, Physical properties, Chemical properties.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Coal, Petroleum and Natural Resources

Topics

- Fossil fuels
- Coal
- Petroleum
- Natural gas
- Conservation
- Renewable/non-renewable resources

Important concepts: Fossil fuels, Coal, Petroleum, Natural gas.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — Force and Pressure

Topics

- Force
- Contact forces
- Non-contact forces
- Pressure
- Pressure and area
- Atmospheric pressure
- Liquid pressure

Important concepts: Force, Contact forces, Non-contact forces, Pressure.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Friction

Topics

- Friction
- Factors affecting friction
- Advantages
- Disadvantages
- Friction reduction/increase

Important concepts: Friction, Factors affecting friction, Advantages, Disadvantages.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Sound

Topics

- Vibrations
- Sound production
- Medium
- Propagation
- Frequency
- Amplitude
- Pitch
- Loudness
- Human hearing

Important concepts: Vibrations, Sound production, Medium, Propagation.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Electricity and Chemical Effects

Topics

- Electric current
- Conductors
- Electrolytes
- Chemical effects
- Electroplating
- Safety

Important concepts: Electric current, Conductors, Electrolytes, Chemical effects.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Light

Topics

- Reflection
- Regular/diffuse reflection
- Plane mirrors
- Multiple reflections
- Refraction as introductory concept
- Human eye

Important concepts: Reflection, Regular/diffuse reflection, Plane mirrors, Multiple reflections.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 10 — Cell Structure

Topics

- Cell as basic unit
- Plant cell
- Animal cell
- Cell membrane
- Cytoplasm
- Nucleus
- Organelles
- Introductory cell division

Important concepts: Cell as basic unit, Plant cell, Animal cell, Cell membrane.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 11 — Reproduction and Adolescence

Topics

- Reproduction
- Puberty
- Adolescence
- Hormonal changes
- Reproductive systems
- Reproductive health

Important concepts: Reproduction, Puberty, Adolescence, Hormonal changes.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 12 — Microorganisms

Topics

- Bacteria
- Fungi
- Protozoa
- Viruses as basic awareness
- Useful microorganisms
- Harmful microorganisms
- Food preservation
- Diseases

Important concepts: Bacteria, Fungi, Protozoa, Viruses as basic awareness.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 13 — Environment and Conservation

Topics

- Ecosystem
- Food chains
- Food webs
- Pollution
- Natural resources
- Conservation
- Sustainable practices

Important concepts: Ecosystem, Food chains, Food webs, Pollution.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

CLASS 9

School-level foundation with progressive Olympiad application.

Mathematics

Focus: computational accuracy, patterns, algebraic thinking, geometry, data interpretation and logical problem solving.

Chapter 1 — Number Systems

Topics

- Rational numbers
- Irrational numbers
- Real numbers
- Number line
- Decimal expansions
- Surds
- Laws of exponents

Important concepts: Rational numbers, Irrational numbers, Real numbers, Number line.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Polynomials

Topics

- Polynomial
- Terms
- Coefficients
- Degree
- Zeroes
- Algebraic operations
- Factorisation
- Identities

Important concepts: Polynomial, Terms, Coefficients, Degree.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Coordinate Geometry

Topics

- Cartesian plane
- Coordinates
- Quadrants
- Plotting points
- Geometric interpretation

Important concepts: Cartesian plane, Coordinates, Quadrants, Plotting points.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Linear Equations in Two Variables

Topics

- Equation formation
- Solutions
- Ordered pairs
- Graphs
- Linear relationships

Important concepts: Equation formation, Solutions, Ordered pairs, Graphs.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — Euclidean Geometry

Topics

- Definitions
- Axioms
- Postulates
- Geometrical reasoning
- Proof-based thinking

Important concepts: Definitions, Axioms, Postulates, Geometrical reasoning.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Lines and Angles

Topics

- Parallel lines
- Transversals
- Angle relationships
- Proofs
- Applications

Important concepts: Parallel lines, Transversals, Angle relationships, Proofs.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Triangles

Topics

- Congruence
- SSS
- SAS
- ASA
- RHS
- Triangle inequalities
- Proof-based problems

Important concepts: Congruence, SSS, SAS, ASA.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Quadrilaterals

Topics

- Properties
- Parallelograms
- Rectangles
- Rhombi
- Squares
- Trapeziums
- Midpoint theorem

Important concepts: Properties, Parallelograms, Rectangles, Rhombi.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Circles

Topics

- Chords
- Arcs
- Angles
- Cyclic properties
- Introductory tangent concepts

Important concepts: Chords, Arcs, Angles, Cyclic properties.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 10 — Mensuration

Topics

- Heron's formula
- Area of triangles
- Quadrilaterals
- Surface area
- Volume
- Composite solids

Important concepts: Heron's formula, Area of triangles, Quadrilaterals, Surface area.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 11 — Statistics

Topics

- Data
- Frequency distribution
- Mean
- Median
- Mode
- Graphical representation
- Interpretation

Important concepts: Data, Frequency distribution, Mean, Median.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 12 — Probability

Topics

- Random experiment
- Outcome
- Sample space
- Event
- Favourable outcomes
- Basic probability

Important concepts: Random experiment, Outcome, Sample space, Event.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 13 — Algebraic Reasoning

Topics

- Identities
- Factorisation
- Algebraic manipulation
- Word problems
- Mathematical modelling

Important concepts: Identities, Factorisation, Algebraic manipulation, Word problems.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 14 — Patterns and Sequences

Topics

- Number sequences
- Arithmetic patterns
- Algebraic patterns
- Generalisation

Important concepts: Number sequences, Arithmetic patterns, Algebraic patterns, Generalisation.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 15 — Mathematical Problem Solving

Topics

- Multi-step problems
- Logical deduction
- Proof
- Data interpretation
- Olympiad reasoning

Important concepts: Multi-step problems, Logical deduction, Proof, Data interpretation.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Science

Focus: conceptual science, observation, cause-and-effect reasoning, diagrams, experiments, applications and interpretation.

Chapter 1 — Matter Around Us

Topics

- States of matter
- Particle model
- Kinetic theory
- Diffusion
- Change of state
- Evaporation
- Temperature and pressure effects

Important concepts: States of matter, Particle model, Kinetic theory, Diffusion.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Pure Substances and Mixtures

Topics

- Elements
- Compounds
- Mixtures
- Homogeneous/heterogeneous mixtures
- Separation techniques
- Solutions
- Suspensions
- Colloids

Important concepts: Elements, Compounds, Mixtures, Homogeneous/heterogeneous mixtures.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Atoms and Molecules

Topics

- Atomic theory
- Atoms
- Molecules
- Atomic mass
- Molecular mass
- Chemical formulae
- Valency
- Introductory mole calculations

Important concepts: Atomic theory, Atoms, Molecules, Atomic mass.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Structure of Atom

Topics

- Electrons
- Protons
- Neutrons
- Atomic number
- Mass number
- Isotopes
- Electronic configuration
- Atomic models

Important concepts: Electrons, Protons, Neutrons, Atomic number.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — Fundamental Unit of Life

Topics

- Cell theory
- Prokaryotic cells
- Eukaryotic cells
- Plant cells
- Animal cells
- Organelles
- Cell membrane
- Nucleus
- Cell division

Important concepts: Cell theory, Prokaryotic cells, Eukaryotic cells, Plant cells.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Tissues

Topics

- Plant tissues
- Meristematic tissues
- Permanent tissues
- Xylem
- Phloem
- Animal tissues
- Epithelial
- Connective
- Muscular
- Nervous

Important concepts: Plant tissues, Meristematic tissues, Permanent tissues, Xylem.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Diversity of Living Organisms

Topics

- Classification
- Taxonomic categories
- Major plant groups
- Major animal groups
- Biodiversity

Important concepts: Classification, Taxonomic categories, Major plant groups, Major animal groups.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Motion

Topics

- Distance
- Displacement
- Speed
- Velocity
- Acceleration
- Graphs
- Uniform/non-uniform motion

Important concepts: Distance, Displacement, Speed, Velocity.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Force and Laws of Motion

Topics

- Force
- Inertia
- Newton's laws
- Momentum
- Action-reaction
- Applications

Important concepts: Force, Inertia, Newton's laws, Momentum.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 10 — Gravitation and Pressure

Topics

- Universal gravitation
- Gravity
- Mass and weight
- Free fall
- Pressure
- Buoyancy
- Archimedes' principle

Important concepts: Universal gravitation, Gravity, Mass and weight, Free fall.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 11 — Work and Energy

Topics

- Work
- Energy
- Kinetic energy
- Potential energy
- Power
- Conservation of energy

Important concepts: Work, Energy, Kinetic energy, Potential energy.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 12 — Sound

Topics

- Production
- Propagation
- Frequency
- Amplitude
- Wavelength
- Speed
- Reflection
- Echo

Important concepts: Production, Propagation, Frequency, Amplitude.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 13 — Environment

Topics

- Ecosystem
- Food chain
- Food web
- Biotic/abiotic factors
- Natural resources
- Pollution
- Conservation

Important concepts: Ecosystem, Food chain, Food web, Biotic/abiotic factors.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

CLASS 10

School-level foundation with progressive Olympiad application.

Mathematics

Focus: computational accuracy, patterns, algebraic thinking, geometry, data interpretation and logical problem solving.

Chapter 1 — Real Numbers

Topics

- Euclid's division concept
- Fundamental theorem of arithmetic
- Prime factorisation
- HCF and LCM
- Irrational numbers
- Rational/irrational relationships

Important concepts: Euclid's division concept, Fundamental theorem of arithmetic, Prime factorisation, HCF and LCM.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Polynomials

Topics

- Degree
- Zeroes
- Graphical interpretation
- Relationship between zeroes and coefficients
- Polynomial factorisation

Important concepts: Degree, Zeroes, Graphical interpretation, Relationship between zeroes and coefficients.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Pair of Linear Equations

Topics

- Two-variable equations
- Graphical method
- Substitution
- Elimination
- Consistency
- Word problems

Important concepts: Two-variable equations, Graphical method, Substitution, Elimination.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Quadratic Equations

Topics

- Standard form
- Factorisation
- Quadratic formula
- Discriminant
- Nature of roots
- Applications

Important concepts: Standard form, Factorisation, Quadratic formula, Discriminant.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — Arithmetic Progressions

Topics

- Sequence
- First term
- Common difference
- nth term
- Sum of n terms
- Applications

Important concepts: Sequence, First term, Common difference, nth term.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Triangles and Similarity

Topics

- Similar figures
- Similar triangles
- Criteria
- Basic proportionality theorem
- Pythagorean theorem
- Converse
- Applications

Important concepts: Similar figures, Similar triangles, Criteria, Basic proportionality theorem.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Coordinate Geometry

Topics

- Distance formula
- Section formula
- Midpoint
- Area of triangle
- Coordinate applications

Important concepts: Distance formula, Section formula, Midpoint, Area of triangle.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Introduction to Trigonometry

Topics

- Trigonometric ratios
- Right triangles
- Standard values
- Identities
- Complementary angles
- Applications

Important concepts: Trigonometric ratios, Right triangles, Standard values, Identities.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Applications of Trigonometry

Topics

- Heights
- Distances
- Angles of elevation
- Angles of depression
- Practical applications

Important concepts: Heights, Distances, Angles of elevation, Angles of depression.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 10 — Circles

Topics

- Tangent
- Radius
- Tangent properties
- Equal tangents
- Chords
- Circle geometry

Important concepts: Tangent, Radius, Tangent properties, Equal tangents.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 11 — Areas Related to Circles

Topics

- Circumference
- Area
- Sector
- Segment
- Arc length
- Composite figures

Important concepts: Circumference, Area, Sector, Segment.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 12 — Surface Areas and Volumes

Topics

- Cubes
- Cuboids
- Cylinders
- Cones
- Spheres
- Hemispheres
- Composite solids
- Conversion problems

Important concepts: Cubes, Cuboids, Cylinders, Cones.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 13 — Statistics

Topics

- Grouped data
- Mean
- Median
- Mode
- Frequency distributions
- Graphical interpretation

Important concepts: Grouped data, Mean, Median, Mode.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 14 — Probability

Topics

- Random experiments
- Sample space
- Events
- Favourable outcomes
- Probability
- Complementary events

Important concepts: Random experiments, Sample space, Events, Favourable outcomes.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 15 — Commercial and Applied Mathematics

Topics

- GST
- Banking applications
- Simple interest
- Compound interest
- Financial applications
- Board-dependent / optional extension

Important concepts: GST, Banking applications, Simple interest, Compound interest.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Science

Focus: conceptual science, observation, cause-and-effect reasoning, diagrams, experiments, applications and interpretation.

Chapter 1 — Chemical Reactions and Equations

Topics

- Chemical reactions
- Chemical equations
- Balancing
- Combination
- Decomposition
- Displacement
- Double displacement
- Oxidation
- Reduction
- Redox
- Corrosion
- Rancidity

Important concepts: Chemical reactions, Chemical equations, Balancing, Combination.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Acids, Bases and Salts

Topics

- Acids
- Bases
- Indicators
- pH
- Neutralisation
- Salts
- Common salts
- Applications

Important concepts: Acids, Bases, Indicators, pH.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Metals and Non-metals

Topics

- Physical properties
- Chemical properties
- Reactivity
- Ionic compounds
- Extraction concepts
- Corrosion
- Alloys

Important concepts: Physical properties, Chemical properties, Reactivity, Ionic compounds.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Carbon and Its Compounds

Topics

- Carbon bonding
- Covalent bonds
- Hydrocarbons
- Functional groups
- Homologous series
- Nomenclature basics
- Chemical properties
- Soaps and detergents

Important concepts: Carbon bonding, Covalent bonds, Hydrocarbons, Functional groups.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — Periodic Classification

Topics

- Modern periodic table
- Atomic number
- Groups
- Periods
- Valency
- Periodic trends
- Metallic/non-metallic character

Important concepts: Modern periodic table, Atomic number, Groups, Periods.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Life Processes

Topics

- Nutrition
- Respiration
- Transportation
- Excretion
- Plants
- Humans

Important concepts: Nutrition, Respiration, Transportation, Excretion.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Control and Coordination

Topics

- Nervous system
- Neurons
- Reflex action
- Brain
- Spinal cord
- Hormones
- Plant coordination

Important concepts: Nervous system, Neurons, Reflex action, Brain.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Reproduction

Topics

- Asexual reproduction
- Sexual reproduction
- Plants
- Human reproduction
- Reproductive health
- Fertilisation
- Development

Important concepts: Asexual reproduction, Sexual reproduction, Plants, Human reproduction.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Heredity and Evolution

Topics

- Traits
- Variation
- Mendelian inheritance
- Genes
- Chromosomes
- Sex determination
- Evolution
- Natural selection

Important concepts: Traits, Variation, Mendelian inheritance, Genes.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 10 — Light: Reflection and Refraction

Topics

- Reflection
- Mirrors
- Mirror formula
- Magnification
- Refraction
- Lenses
- Lens formula
- Power of lens

Important concepts: Reflection, Mirrors, Mirror formula, Magnification.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 11 — Human Eye and Optical Phenomena

Topics

- Human eye
- Accommodation
- Vision defects
- Correction
- Dispersion
- Atmospheric effects

Important concepts: Human eye, Accommodation, Vision defects, Correction.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 12 — Electricity

Topics

- Charge
- Current
- Potential difference
- Resistance
- Ohm's law
- Series/parallel circuits
- Electrical power
- Electrical energy

Important concepts: Charge, Current, Potential difference, Resistance.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 13 — Magnetism and Electromagnetism

Topics

- Magnetic field
- Field due to current
- Electromagnets
- Fleming's rules
- Electromagnetic induction
- Electric motor
- Generator
- Domestic electricity

Important concepts: Magnetic field, Field due to current, Electromagnets, Fleming's rules.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

CLASS 11

Senior-secondary subject-separated syllabus. Questions remain anchored to the concepts normally taught at this stage.

Mathematics

Focus: computational accuracy, patterns, algebraic thinking, geometry, data interpretation and logical problem solving.

Chapter 1 — Sets

Topics

- Sets
- Representation
- Empty set
- Finite/infinite sets
- Subsets
- Universal set
- Venn diagrams
- Union
- Intersection
- Difference
- Complement

Important concepts: Sets, Representation, Empty set, Finite/infinite sets.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Relations and Functions

Topics

- Ordered pairs
- Cartesian product
- Relations
- Domain
- Codomain
- Range
- Functions
- Types
- Function notation
- Composition
- Graphs

Important concepts: Ordered pairs, Cartesian product, Relations, Domain.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Trigonometric Functions

Topics

- Angle measurement
- Degrees/radians
- Trigonometric ratios
- Unit circle
- Signs
- Standard values
- Identities
- Compound angles
- Trigonometric equations

Important concepts: Angle measurement, Degrees/radians, Trigonometric ratios, Unit circle.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Complex Numbers and Quadratic Equations

Topics

- Imaginary unit
- Complex numbers
- Real/imaginary parts
- Algebra
- Argand plane
- Modulus
- Conjugate
- Quadratic equations
- Roots
- Nature of roots

Important concepts: Imaginary unit, Complex numbers, Real/imaginary parts, Algebra.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — Linear Inequalities

Topics

- Inequalities
- One variable
- Number-line representation
- Two-variable inequalities
- Graphical solution
- Systems

Important concepts: Inequalities, One variable, Number-line representation, Two-variable inequalities.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Permutations and Combinations

Topics

- Fundamental counting principle
- Factorial
- Permutations
- Combinations
- nPr
- nCr
- Applications

Important concepts: Fundamental counting principle, Factorial, Permutations, Combinations.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Sequences and Series

Topics

- Sequences
- Series
- Arithmetic progression
- Geometric progression
- n th term
- Sum
- AM/GM

Important concepts: Sequences, Series, Arithmetic progression, Geometric progression.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Straight Lines

Topics

- Coordinate plane
- Distance
- Slope
- Angle between lines
- Point-slope form
- Two-point form
- Intercept form
- General equation
- Parallel/perpendicular lines

Important concepts: Coordinate plane, Distance, Slope, Angle between lines.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Conic Sections

Topics

- Circle
- Parabola
- Ellipse
- Hyperbola
- Standard equations
- Basic geometric properties

Important concepts: Circle, Parabola, Ellipse, Hyperbola.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 10 — Introduction to 3-D Geometry

Topics

- Coordinates
- Coordinate axes
- Coordinate planes
- 3-D points
- Distance between points

Important concepts: Coordinates, Coordinate axes, Coordinate planes, 3-D points.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 11 — Limits and Derivatives

Topics

- Intuitive limit
- Algebraic limits
- Derivative
- Rate of change
- Geometric meaning
- Basic differentiation
- Standard derivatives

Important concepts: Intuitive limit, Algebraic limits, Derivative, Rate of change.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 12 — Statistics

Topics

- Data
- Frequency distributions
- Mean
- Median
- Mode
- Range
- Variance
- Standard deviation

Important concepts: Data, Frequency distributions, Mean, Median.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 13 — Probability

Topics

- Random experiment
- Sample space
- Events
- Mutually exclusive events
- Exhaustive events
- Probability
- Basic addition principles

Important concepts: Random experiment, Sample space, Events, Mutually exclusive events.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Physics

Focus: physical principles, mathematical modelling, graphs, numerical reasoning, experimental interpretation and multi-step applications.

Chapter 1 — Physical World, Units and Measurement

Topics

- Physical quantities
- SI units
- Dimensions
- Significant figures
- Measurement errors
- Vernier calipers
- Screw gauge
- Experimental reasoning

Important concepts: Physical quantities, SI units, Dimensions, Significant figures.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Motion in One and Two Dimensions

Topics

- Distance
- Displacement
- Speed
- Velocity
- Acceleration
- Motion graphs
- Equations of motion
- Projectile motion
- Relative motion

Important concepts: Distance, Displacement, Speed, Velocity.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Laws of Motion

Topics

- Force
- Inertia
- Newton's laws
- Momentum
- Friction
- Circular motion
- Applications

Important concepts: Force, Inertia, Newton's laws, Momentum.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Work, Energy and Power

Topics

- Work
- Energy
- Kinetic energy
- Potential energy
- Work-energy theorem
- Conservation of energy
- Power

Important concepts: Work, Energy, Kinetic energy, Potential energy.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — System of Particles and Rotational Motion

Topics

- Centre of mass
- Momentum
- Torque
- Angular momentum
- Moment of inertia
- Rotational equilibrium

Important concepts: Centre of mass, Momentum, Torque, Angular momentum.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Gravitation

Topics

- Universal law
- Gravitational field
- Potential
- Acceleration due to gravity
- Satellites
- Escape velocity
- Orbital motion

Important concepts: Universal law, Gravitational field, Potential, Acceleration due to gravity.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Properties of Bulk Matter

Topics

- Elasticity
- Fluids
- Pressure
- Viscosity
- Surface tension
- Bernoulli principle
- Thermal expansion

Important concepts: Elasticity, Fluids, Pressure, Viscosity.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Thermodynamics and Kinetic Theory

Topics

- Temperature
- Heat
- Thermodynamic systems
- First law
- Second-law concepts
- Gas laws
- Kinetic theory
- Molecular motion

Important concepts: Temperature, Heat, Thermodynamic systems, First law.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Oscillations

Topics

- Periodic motion
- Simple harmonic motion
- Amplitude
- Time period
- Frequency
- Energy in SHM

Important concepts: Periodic motion, Simple harmonic motion, Amplitude, Time period.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 10 — Waves

Topics

- Wave motion
- Transverse/longitudinal waves
- Wavelength
- Frequency
- Speed
- Sound
- Superposition
- Standing waves

Important concepts: Wave motion, Transverse/longitudinal waves, Wavelength, Frequency.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chemistry

Focus: concepts, equations, structure-property relationships, calculations, reaction reasoning and interpretation of chemical situations.

Chapter 1 — Some Basic Concepts of Chemistry

Topics

- Matter
- Laws of chemical combination
- Atomic/molecular masses
- Mole concept
- Molar mass
- Stoichiometry
- Percentage composition
- Limiting reagent

Important concepts: Matter, Laws of chemical combination, Atomic/molecular masses, Mole concept.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Structure of Atom

Topics

- Subatomic particles
- Atomic models
- Electromagnetic radiation
- Quantum ideas
- Orbitals
- Electronic configuration

Important concepts: Subatomic particles, Atomic models, Electromagnetic radiation, Quantum ideas.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Classification of Elements and Periodicity

Topics

- Modern periodic law
- Periodic table
- Atomic radius
- Ionisation energy
- Electron affinity
- Electronegativity
- Valency
- Periodic trends

Important concepts: Modern periodic law, Periodic table, Atomic radius, Ionisation energy.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Chemical Bonding

Topics

- Ionic bonding
- Covalent bonding
- Lewis structures
- VSEPR
- Hybridisation
- Molecular geometry
- Polarity
- Hydrogen bonding
- Coordinate bonding

Important concepts: Ionic bonding, Covalent bonding, Lewis structures, VSEPR.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — States of Matter

Topics

- Gas laws
- Ideal gas equation
- Kinetic theory
- Liquids
- Vapour pressure
- Intermolecular forces

Important concepts: Gas laws, Ideal gas equation, Kinetic theory, Liquids.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Thermodynamics

Topics

- System and surroundings
- Internal energy
- Enthalpy
- Heat
- Work
- Hess's law
- Thermochemical equations

Important concepts: System and surroundings, Internal energy, Enthalpy, Heat.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Equilibrium

Topics

- Physical equilibrium
- Chemical equilibrium
- Le Chatelier principle
- Equilibrium constant
- Ionic equilibrium
- Acids/bases
- pH
- Buffer concepts

Important concepts: Physical equilibrium, Chemical equilibrium, Le Chatelier principle, Equilibrium constant.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Redox Reactions

Topics

- Oxidation
- Reduction
- Oxidation number
- Redox equations
- Balancing redox reactions

Important concepts: Oxidation, Reduction, Oxidation number, Redox equations.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Organic Chemistry Fundamentals

Topics

- Organic compounds
- Classification
- Nomenclature
- Isomerism basics
- Electronic effects
- Reaction intermediates
- Hydrocarbons

Important concepts: Organic compounds, Classification, Nomenclature, Isomerism basics.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Biology

Focus: biological structures and processes, relationships, diagrams, experimental reasoning, genetics, physiology and ecology.

Chapter 1 — The Living World

Topics

- Characteristics of life
- Biodiversity
- Taxonomy
- Systematics
- Binomial nomenclature
- Taxonomic hierarchy
- Taxonomic aids

Important concepts: Characteristics of life, Biodiversity, Taxonomy, Systematics.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Biological Classification

Topics

- Kingdom systems
- Monera
- Protista
- Fungi
- Plantae
- Animalia
- Viruses
- Viroids
- Lichens

Important concepts: Kingdom systems, Monera, Protista, Fungi.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Plant Kingdom

Topics

- Algae
- Bryophytes
- Pteridophytes
- Gymnosperms
- Angiosperms
- Monocots/dicots
- Life cycles
- Alternation of generations

Important concepts: Algae, Bryophytes, Pteridophytes, Gymnosperms.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Animal Kingdom

Topics

- Animal classification
- Symmetry
- Body organisation
- Coelom
- Segmentation
- Notochord
- Major animal phyla
- Chordates

Important concepts: Animal classification, Symmetry, Body organisation, Coelom.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — Morphology and Anatomy of Flowering Plants

Topics

- Root
- Stem
- Leaf
- Inflorescence
- Flower
- Fruit
- Seed
- Plant tissues
- Vascular systems
- Monocot/dicot structures

Important concepts: Root, Stem, Leaf, Inflorescence.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Structural Organisation in Animals

Topics

- Animal tissues
- Epithelial tissue
- Connective tissue
- Muscular tissue
- Nervous tissue
- Basic animal anatomy

Important concepts: Animal tissues, Epithelial tissue, Connective tissue, Muscular tissue.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Cell Biology and Biomolecules

Topics

- Cell theory
- Prokaryotic/eukaryotic cells
- Cell organelles
- Membranes
- Carbohydrates
- Proteins
- Lipids
- Nucleic acids
- Enzymes

Important concepts: Cell theory, Prokaryotic/eukaryotic cells, Cell organelles, Membranes.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Cell Cycle and Cell Division

Topics

- Cell cycle
- Interphase
- Mitosis
- Meiosis
- Crossing over
- Genetic variation

Important concepts: Cell cycle, Interphase, Mitosis, Meiosis.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Plant Physiology

Topics

- Mineral nutrition
- Photosynthesis
- Respiration
- Plant growth
- Plant hormones
- Photoperiodism
- Vernalisation

Important concepts: Mineral nutrition, Photosynthesis, Respiration, Plant growth.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 10 — Human Physiology

Topics

- Digestion
- Breathing
- Circulation
- Excretion
- Locomotion
- Neural coordination
- Endocrine coordination

Important concepts: Digestion, Breathing, Circulation, Excretion.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

CLASS 12

Senior-secondary subject-separated syllabus. Questions remain anchored to the concepts normally taught at this stage.

Mathematics

Focus: computational accuracy, patterns, algebraic thinking, geometry, data interpretation and logical problem solving.

Chapter 1 — Relations and Functions

Topics

- Relations
- Functions
- Types
- Composition
- Invertible functions
- Binary operations

Important concepts: Relations, Functions, Types, Composition.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Inverse Trigonometric Functions

Topics

- Principal values
- Domains and ranges
- Properties
- Identities
- Graphs
- Applications

Important concepts: Principal values, Domains and ranges, Properties, Identities.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Matrices

Topics

- Types
- Order
- Elements
- Equality
- Addition
- Multiplication
- Transpose
- Inverse
- Applications

Important concepts: Types, Order, Elements, Equality.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Determinants

Topics

- Determinants
- Properties
- Minors
- Cofactors
- Area applications
- Solving equations

Important concepts: Determinants, Properties, Minors, Cofactors.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — Continuity and Differentiability

Topics

- Continuity
- Limits
- Differentiability
- Derivatives
- Chain rule
- Implicit differentiation
- Related concepts

Important concepts: Continuity, Limits, Differentiability, Derivatives.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Applications of Derivatives

Topics

- Rate of change
- Increasing/decreasing functions
- Maxima/minima
- Tangents
- Normals
- Optimisation

Important concepts: Rate of change, Increasing/decreasing functions, Maxima/minima, Tangents.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Integrals

Topics

- Indefinite integrals
- Standard integrals
- Substitution
- Integration by parts
- Partial fractions

Important concepts: Indefinite integrals, Standard integrals, Substitution, Integration by parts.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Applications of Integrals

Topics

- Area under curves
- Area between curves
- Geometric interpretation

Important concepts: Area under curves, Area between curves, Geometric interpretation.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Differential Equations

Topics

- Order and degree
- Formation
- Variable separable equations
- General and particular solutions
- Applications

Important concepts: Order and degree, Formation, Variable separable equations, General and particular solutions.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 10 — Vector Algebra

Topics

- Vectors
- Magnitude
- Direction
- Components
- Dot product
- Cross product
- Geometric applications

Important concepts: Vectors, Magnitude, Direction, Components.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 11 — Three-Dimensional Geometry

Topics

- Direction ratios
- Direction cosines
- Lines
- Planes
- Angles
- Distances

Important concepts: Direction ratios, Direction cosines, Lines, Planes.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 12 — Linear Programming

Topics

- Constraints
- Objective function
- Feasible region
- Graphical method
- Optimisation

Important concepts: Constraints, Objective function, Feasible region, Graphical method.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 13 — Probability

Topics

- Conditional probability
- Independent events
- Bayes' theorem
- Random variables
- Probability distributions
- Binomial distribution

Important concepts: Conditional probability, Independent events, Bayes' theorem, Random variables.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Physics

Focus: physical principles, mathematical modelling, graphs, numerical reasoning, experimental interpretation and multi-step applications.

Chapter 1 — Electrostatics

Topics

- Electric charge
- Coulomb's law
- Electric field
- Field lines
- Electric potential
- Potential difference
- Dipoles
- Capacitance
- Capacitors
- Energy stored

Important concepts: Electric charge, Coulomb's law, Electric field, Field lines.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Current Electricity

Topics

- Electric current
- Drift
- Resistance
- Resistivity
- Ohm's law
- Series/parallel circuits
- Kirchhoff's laws
- Cells
- Internal resistance
- Electrical power

Important concepts: Electric current, Drift, Resistance, Resistivity.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Moving Charges and Magnetism

Topics

- Magnetic field
- Lorentz force
- Force on current-carrying conductor
- Biot-Savart law
- Ampere's law
- Motion in magnetic fields
- Cyclotron concept

Important concepts: Magnetic field, Lorentz force, Force on current-carrying conductor, Biot-Savart law.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Magnetism and Matter

Topics

- Magnetic materials
- Bar magnet
- Earth's magnetism
- Magnetic field
- Magnetic properties

Important concepts: Magnetic materials, Bar magnet, Earth's magnetism, Magnetic field.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — Electromagnetic Induction and Alternating Current

Topics

- Faraday's law
- Lenz's law
- Motional EMF
- Self/mutual induction
- AC
- RMS values
- Reactance
- Impedance
- Transformer

Important concepts: Faraday's law, Lenz's law, Motional EMF, Self/mutual induction.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Electromagnetic Waves

Topics

- Electromagnetic spectrum
- Properties
- Applications
- Wave propagation

Important concepts: Electromagnetic spectrum, Properties, Applications, Wave propagation.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Ray and Wave Optics

Topics

- Reflection
- Refraction
- Mirrors
- Lenses
- Optical instruments
- Wavefront
- Interference
- Diffraction
- Polarisation

Important concepts: Reflection, Refraction, Mirrors, Lenses.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Dual Nature, Atoms and Nuclei

Topics

- Photoelectric effect
- Matter waves
- Atomic models
- Bohr model
- Nuclear structure
- Radioactivity
- Nuclear reactions
- Fission/fusion

Important concepts: Photoelectric effect, Matter waves, Atomic models, Bohr model.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Semiconductor Electronics

Topics

- Semiconductors
- Intrinsic/extrinsic
- p-n junction
- Diode
- Rectification
- Transistor
- Logic gates
- Digital applications

Important concepts: Semiconductors, Intrinsic/extrinsic, p-n junction, Diode.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chemistry

Focus: concepts, equations, structure-property relationships, calculations, reaction reasoning and interpretation of chemical situations.

Chapter 1 — Solutions

Topics

- Types of solutions
- Concentration
- Solubility
- Molarity
- Molality
- Colligative properties
- Osmosis

Important concepts: Types of solutions, Concentration, Solubility, Molarity.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Electrochemistry

Topics

- Electrolytes
- Electrochemical cells
- Electrode potential
- Nernst equation
- Conductance
- Electrolysis
- Batteries
- Corrosion

Important concepts: Electrolytes, Electrochemical cells, Electrode potential, Nernst equation.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Chemical Kinetics

Topics

- Reaction rate
- Factors affecting rate
- Rate law
- Order
- Molecularity
- Activation energy
- Arrhenius equation
- Half-life

Important concepts: Reaction rate, Factors affecting rate, Rate law, Order.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Surface Chemistry

Topics

- Adsorption
- Catalysis
- Colloids
- Emulsions
- Surface phenomena

Important concepts: Adsorption, Catalysis, Colloids, Emulsions.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — d- and f-Block Elements

Topics

- Transition elements
- Electronic configuration
- Oxidation states
- Colour
- Magnetic properties
- Lanthanides
- Actinides

Important concepts: Transition elements, Electronic configuration, Oxidation states, Colour.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Coordination Compounds

Topics

- Coordination entities
- Ligands
- Coordination number
- Nomenclature
- Isomerism
- Bonding
- Applications

Important concepts: Coordination entities, Ligands, Coordination number, Nomenclature.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Haloalkanes, Haloarenes and Alcohols

Topics

- Nomenclature
- Preparation
- Properties
- Substitution
- Elimination
- Alcohols
- Phenols
- Ethers

Important concepts: Nomenclature, Preparation, Properties, Substitution.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Aldehydes, Ketones, Carboxylic Acids and Amines

Topics

- Functional groups
- Preparation
- Properties
- Reactions
- Oxidation/reduction
- Amines
- Basicity

Important concepts: Functional groups, Preparation, Properties, Reactions.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Biomolecules and Polymers

Topics

- Carbohydrates
- Proteins
- Amino acids
- Enzymes
- Vitamins
- Nucleic acids
- Polymers
- Everyday chemistry

Important concepts: Carbohydrates, Proteins, Amino acids, Enzymes.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 10 — Chemistry in Everyday Life

Topics

- Medicines
- Drugs
- Soaps
- Detergents
- Food-related chemicals
- Environmental applications

Important concepts: Medicines, Drugs, Soaps, Detergents.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Biology

Focus: biological structures and processes, relationships, diagrams, experimental reasoning, genetics, physiology and ecology.

Chapter 1 — Reproduction in Organisms

Topics

- Reproductive strategies
- Asexual reproduction
- Sexual reproduction
- Life cycles

Important concepts: Reproductive strategies, Asexual reproduction, Sexual reproduction, Life cycles.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 2 — Sexual Reproduction in Flowering Plants

Topics

- Flower structure
- Gametogenesis
- Pollination
- Fertilisation
- Double fertilisation
- Seed
- Fruit
- Embryo

Important concepts: Flower structure, Gametogenesis, Pollination, Fertilisation.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 3 — Human Reproduction

Topics

- Male reproductive system
- Female reproductive system
- Gametogenesis
- Menstrual cycle
- Fertilisation
- Implantation
- Pregnancy
- Parturition
- Lactation

Important concepts: Male reproductive system, Female reproductive system, Gametogenesis, Menstrual cycle.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 4 — Reproductive Health

Topics

- Reproductive health
- Contraception
- STIs
- Infertility
- Assisted reproductive technologies
- Awareness and prevention

Important concepts: Reproductive health, Contraception, STIs, Infertility.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 5 — Principles of Inheritance

Topics

- Mendelian genetics
- Laws of inheritance
- Monohybrid/dihybrid crosses
- Dominance
- Codominance
- Multiple alleles
- Sex-linked inheritance
- Pedigree concepts

Important concepts: Mendelian genetics, Laws of inheritance, Monohybrid/dihybrid crosses, Dominance.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 6 — Molecular Basis of Inheritance

Topics

- DNA
- RNA
- Replication
- Transcription
- Translation
- Genetic code
- Gene expression
- Mutations

Important concepts: DNA, RNA, Replication, Transcription.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 7 — Evolution

Topics

- Origin of life
- Variation
- Natural selection
- Evidence of evolution
- Human evolution
- Population genetics concepts

Important concepts: Origin of life, Variation, Natural selection, Evidence of evolution.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 8 — Human Health and Disease

Topics

- Pathogens
- Immunity
- Vaccination
- Infectious diseases
- Non-infectious diseases
- Cancer
- AIDS
- Public-health principles

Important concepts: Pathogens, Immunity, Vaccination, Infectious diseases.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 9 — Biotechnology

Topics

- Biotechnology principles
- Recombinant DNA
- Genetic engineering
- PCR
- Cloning
- Transgenic organisms
- Applications
- Biosafety

Important concepts: Biotechnology principles, Recombinant DNA, Genetic engineering, PCR.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.

Chapter 10 — Ecology and Environment

Topics

- Organisms and populations
- Ecosystems
- Food chains
- Food webs
- Energy flow
- Ecological pyramids
- Biodiversity
- Conservation
- Environmental issues

Important concepts: Organisms and populations, Ecosystems, Food chains, Food webs.

Olympiad application: Questions may test reasoning, comparison, multi-step application, interpretation of diagrams/graphs/tables, error detection, prediction or real-world use of the listed concepts.